Prabhavit Gupta

Software Engineer | Agent Harness, Real-Time AI Systems & Execution Runtimes

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Executive Summary

- **Agent Harnesses & Execution Infrastructure:** Software engineer with 3 years commercial software experience specializing in autonomous agent execution runtimes, tool invocation pipelines, and low-latency LLM interaction systems.
- **Real-Time Systems, Latency & Tool Calling:** Hands-on background developing streaming API connectors, WebSocket handlers, stateful session management, Model Context Protocol (MCP) tool loops, and isolating latency bottlenecks across multi-step agent interactions.
- **Automated Evaluation & High-Agency Execution:** Experienced in building automated evaluation suites (LLM-as-judge scoring, regression tracking with Langfuse and DeepEval), writing resilient integration harnesses in TypeScript and Python, and shipping production capabilities at high velocity.

Technical Skills & Core Competencies

- **Agent Systems & Execution Harnesses:** Agent Tool Loops, Tool Calling, Agent Execution Runtimes, Model Context Protocol (MCP), Sandboxed Execution, Streaming State Management, Prompt Engineering, Prompt-to-Action Validation.
- **AI Evaluation & Observability:** LLM-as-Judge Evals, Regression Suites, Distributed Tracing (Langfuse), DeepEval, Latency Benchmarking, Structured Outputs (Pydantic, Zod).
- **Core Engineering:** TypeScript, Node.js, Python (AsyncIO, WebSockets, PyTest), Next.js, Laravel, REST APIs, GraphQL, SQL (PostgreSQL, SQLite), Docker, Git.
- **Developer Operations & Tooling:** Postman, GitHub Actions, CI/CD, Linux CLI, Agile/Jira, Technical Documentation, Performance Optimization.

Professional Experience

AI Systems Engineer & Agent Infrastructure Consultant • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Architected sandboxed agent execution runtimes connecting LLMs to external APIs, ensuring deterministic tool dispatch, stateful session recovery, and structured schema enforcement.
- Engineered Model Context Protocol (MCP) server and client harnesses, allowing autonomous agents to dynamically query operational endpoints with minimal overhead.
- Established automated evaluation pipelines using Langfuse and LLM-as-judge scoring, tracking tool dispatch accuracy, schema adherence, and end-to-end execution latency across regression suites.
- Implemented streaming WebSocket handlers and asynchronous Python workers to minimize end-to-end response latency for conversational AI workflows.
- Authored technical architecture specs, integration documentation, and automated test harnesses for external developer teams.

Lead Developer & Product Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Scaled platform from 2,000 to 4,500+ active users, architecting core platform features, real-time API integrations, and conversational customer-facing agents.
- Migrated core application architecture off legacy infrastructure onto Next.js and Laravel, ensuring continuous data integrity and zero customer downtime.
- Deployed automated conversational support agents with deterministic boundary guards and human-in-the-loop escalation paths, decreasing ticket resolution time.
- Built automated data ingestion and extraction pipelines aggregating unstructured venue data across multiple European platforms with strict validation checks.
- Integrated external third-party management APIs (OfficeRnD, Nexudus) for automated booking synchronization via webhooks.

Full-Stack Solutions Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Delivered custom API integrations and automated services in Python and TypeScript, connecting client platforms with internal data warehouses.
- Built automated unit and integration testing suites within Docker CI/CD pipelines, increasing software delivery reliability.
- Collaborated with technical and non-technical stakeholders to translate complex domain workflows into clear engi-

neering specifications and production code.

Key Engineering Projects

- **Autonomous Tool-Calling & Evaluation Harness (TypeScript + MCP + Langfuse):** Architected an execution harness enabling LLMs to safely invoke external tools, validating payloads against strict schemas with automated tracing and regression metrics.
- **Universal Multi-Source Ingestion Pipeline (Python + Playwright + REST APIs):** Designed an automated data harvesting and enrichment system parsing 900+ structured and semi-structured external sources with schema validation and error isolation.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to international relocation to London or San Francisco with visa sponsorship; available immediately.